

(1) ETANOL

C2H6O

(2) WATER

H2O

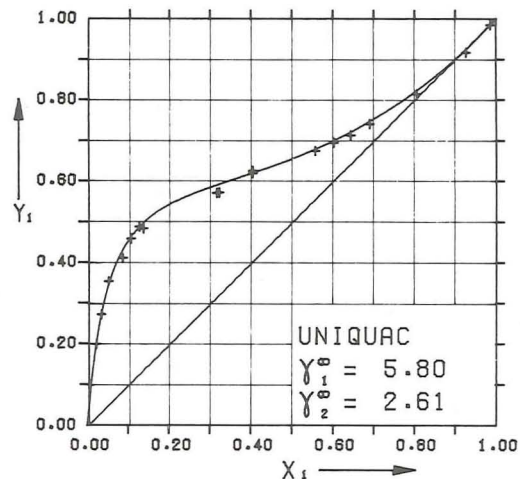
+++++ ANTOINE CONSTANTS REGION +++++ CCNSISTENCY
 (1) 8.11220 1592.864 226.184 20- 93 C METHCD 1 +
 (2) 8.07131 1730.630 233.426 1- 100 C METHCD 2 +

PRESSURE= 760.00 MM HG (1.013 BAR)

LIT: BLOOM C.H., CLUMP C.W., KOECKERT A.H., IND. ENG. CHEM. 53, 829 (1961).

CONSTANTS: A12 A21 ALPHA12

MARGULES	1.6711	0.9006	
VAN LAAR	1.7740	0.5561	
WILSON	421.9205	546.8418	
NRTL	-105.7202	1384.2083	0.2976
UNIQUAC	9.5115	307.7309	



EXPERIMENTAL DATA			MARGULES		VAN LAAR		WILSON		NRTL		UNIQUAC	
T DEG C	X1	Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1
94.30	0.0210	0.1990	-0.54	0.0111	-0.16	0.0002	0.06	-0.0058	-0.30	0.0043	-0.20	0.0013
91.90	0.0330	0.2720	-0.78	0.0135	-0.38	0.0031	-0.20	-0.0011	-0.49	0.0061	-0.41	0.0039
90.00	0.0500	0.3530	-0.28	0.0205	0.05	0.0132	0.13	0.0129	0.01	0.0143	0.05	0.0135
87.30	0.0850	0.4110	0.41	-0.0185	0.49	-0.0179	0.32	-0.0113	0.56	-0.0194	0.50	-0.0180
86.10	0.1050	0.4580	0.54	-0.0077	0.46	-0.0032	0.20	0.0060	0.58	-0.0054	0.49	-0.0033
85.20	0.1250	0.4880	0.68	-0.0055	0.47	0.0022	0.14	0.0129	0.61	-0.0002	0.50	0.0022
84.70	0.1350	0.4840	0.61	-0.0210	0.35	-0.0121	-0.01	-0.0009	0.49	-0.0145	0.37	-0.0120
81.80	0.3150	0.5710	1.17	-0.0310	0.66	-0.0192	0.47	-0.0147	0.74	-0.0184	0.67	-0.0182
31.60	0.3210	0.5720	1.03	-0.0318	0.52	-0.0203	0.34	-0.0161	0.60	-0.0193	0.53	-0.0193
80.60	0.4030	0.6190	0.62	-0.0070	0.23	-0.0011	0.18	-0.0014	0.29	0.0012	0.25	-0.0000
80.20	0.4030	0.6250	0.22	-0.0010	-0.17	0.0049	-0.22	0.0046	-0.11	0.0071	-0.15	0.0060
79.50	0.5560	0.6750	0.39	0.0010	0.21	-0.0039	0.29	-0.0082	0.24	-0.0007	0.23	-0.0030
79.20	0.6020	0.6950	0.32	0.0020	0.18	-0.0050	0.28	-0.0092	0.20	-0.0019	0.20	-0.0042
79.10	0.6430	0.7130	0.41	0.0007	0.30	-0.0076	0.41	-0.0114	0.32	-0.0047	0.32	-0.0068
78.60	0.6890	0.7410	0.09	0.0039	0.02	-0.0048	0.14	-0.0077	0.03	-0.0023	0.04	-0.0041
78.30	0.8050	0.8140	0.08	-0.0018	0.09	-0.0074	0.22	-0.0070	0.09	-0.0062	0.11	-0.0068
78.30	0.9260	0.9170	0.11	-0.0065	0.16	-0.0060	0.24	-0.0037	0.15	-0.0060	0.17	-0.0057
78.20	0.9870	0.9850	-0.08	-0.0011	-0.06	-0.0005	-0.04	0.0003	-0.06	-0.0006	-0.06	-0.0004

MEAN DEVIATION:

0.46 0.0103

0.28 0.0074

0.22 0.0075

0.33 0.0074

0.29 0.0071

MAX. DEVIATION:

1.17 0.0318

0.66 0.0203

0.47 0.0161

0.74 0.0194

0.67 0.0193

(1) ETHANOL

 C_2H_6O

(2) WATER

 H_2O

+++++ ANTOINE CONSTANTS REGION +++++ CONSISTENCY

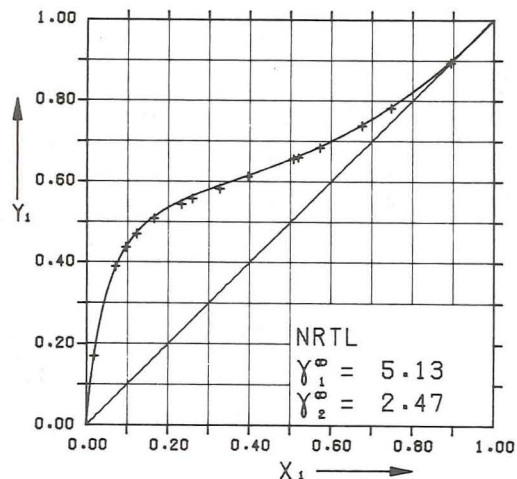
(1)	8.11220	1592.864	226.184	20-	93 C	METHOD 1	+
(2)	8.07131	1730.630	233.426	1-	100 C	METHOD 2	+

PRESSURE= 760.00 MM HG (1.013 BAR)

LIT: CAREY J.S., LEWIS W.K., IND. ENG. CHEM. 24, 882 (1932).

CONSTANTS: A12 A21 ALPHA12

MARGULES	1.6022	0.7947	
VAN LAAR	1.6798	0.9227	
WILSON	325.0757	953.2792	
NRTL	-122.0876	1341.3195	0.3008
UNIQUAC	21.7366	262.9622	



EXPERIMENTAL DATA			MARGULES		VAN LAAR		WILSON		NRTL		UNIQUAC	
T DEG C	X1	Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1
95.50	0.0190	0.1700	-0.05	0.0055	0.20	-0.0022	0.32	-0.0055	0.10	0.0008	0.19	-0.0017
89.00	0.0721	0.3891	0.39	0.0046	0.52	0.0027	0.47	0.0058	0.57	0.0018	0.54	0.0025
86.70	0.0966	0.4375	-0.10	0.0013	-0.09	0.0030	-0.23	0.0085	-0.00	0.0014	-0.06	0.0029
85.30	0.1238	0.4704	-0.04	-0.0065	-0.14	-0.0019	-0.34	0.0052	-0.03	-0.0035	-0.12	-0.0017
84.10	0.1661	0.5089	0.32	-0.0109	0.09	-0.0035	-0.13	0.0039	0.19	-0.0045	0.10	-0.0029
82.70	0.2337	0.5445	0.38	-0.0169	0.10	-0.0091	-0.06	-0.0037	0.17	-0.0085	0.10	-0.0080
82.30	0.2608	0.5580	0.37	-0.0152	0.10	-0.0082	-0.02	-0.0041	0.16	-0.0070	0.10	-0.0070
81.50	0.3273	0.5826	0.28	-0.0141	0.09	-0.0102	0.08	-0.0094	0.11	-0.0078	0.09	-0.0089
80.70	0.3965	0.6122	0.02	-0.0057	-0.05	-0.0062	0.03	-0.0081	-0.05	-0.0029	-0.05	-0.0049
79.80	0.5079	0.6564	-0.14	0.0015	-0.07	-0.0053	0.09	-0.0094	-0.09	-0.0015	-0.06	-0.0042
79.70	0.5198	0.6599	-0.17	0.0005	-0.09	-0.0068	0.09	-0.0109	-0.11	-0.0030	-0.08	-0.0057
79.30	0.5732	0.6841	-0.26	0.0026	-0.13	-0.0063	0.06	-0.0103	-0.16	-0.0028	-0.12	-0.0053
78.74	0.6763	0.7385	-0.30	0.0038	-0.10	-0.0046	0.11	-0.0066	-0.14	-0.0022	-0.09	-0.0039
78.41	0.7472	0.7815	-0.35	0.0010	-0.12	-0.0044	0.08	-0.0043	-0.16	-0.0031	-0.11	-0.0039
78.15	0.8943	0.8943	-0.26	-0.0057	-0.06	-0.0023	0.09	0.0011	-0.09	-0.0029	-0.05	-0.0021
MEAN DEVIATION:			0.23	0.0064	0.13	0.0051	0.15	0.0065	0.14	0.0036	0.13	0.0044
MAX. DEVIATION:			0.39	0.0169	0.52	0.0103	0.47	0.0109	0.57	0.0085	0.54	0.0089

(1) ETHANOL

C2H6O

(2) WATER

H2O

+++++ ANTOINE CONSTANTS REGION +++++

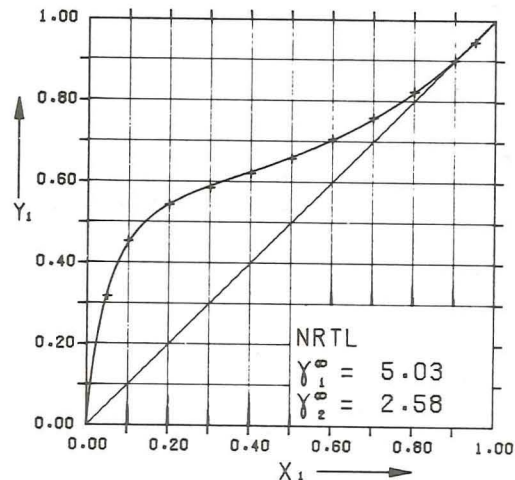
	ANTOINE CONSTANTS		REGION		CONSISTENCY	
(1)	8.11220	1592.864	226.184	20- 93 C	METHOD 1	+
(2)	8.07131	1730.630	233.426	1- 100 C	METHOD 2	+

PRESSURE= 500.00 MM HG (0.667 BAR)

LIT: KIRSCHBAUM E., GERSTNER H., Z.VDI BEIH.VERFAHRENSTECHNIK 1,10(1939).

CONSTANTS: A12 A21 ALPHA12

	A12	A21	ALPHA12
MARGULES	1.5783	0.8971	
VAN LAAR	1.6715	0.9559	
WILSON	333.4674	922.6100	
NRTL	-75.6857	1239.6528	0.2983
UNIQUAC	77.5685	199.4573	



EXPERIMENTAL DATA			MARGULES		VAN LAAR		WILSON		NRTL		UNIQUAC	
T DEG C	X1	Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1
79.60	0.0500	0.3180	-0.53	-0.0011	-0.19	-0.0094	-0.06	-0.0115	-0.30	-0.0067	-0.20	-0.0090
75.70	0.1000	0.4530	-0.32	0.0066	-0.23	0.0066	-0.37	0.0132	-0.22	0.0064	-0.24	0.0073
72.55	0.2000	0.5430	0.16	-0.0094	-0.09	-0.0006	-0.42	0.0102	-0.03	-0.0010	-0.12	0.0009
71.05	0.3000	0.5850	0.15	-0.0141	-0.14	-0.0054	-0.38	0.0015	-0.10	-0.0046	-0.17	-0.0039
70.25	0.4000	0.6210	0.19	-0.0099	-0.02	-0.0056	-0.12	-0.0036	0.00	-0.0037	-0.03	-0.0043
69.60	0.5000	0.6580	0.17	-0.0045	0.07	-0.0055	0.05	-0.0071	0.08	-0.0032	0.06	-0.0047
69.05	0.6000	0.7060	0.17	0.0046	0.13	-0.0003	0.17	-0.0033	0.14	0.0020	0.14	0.0002
68.55	0.7000	0.7600	0.11	0.0079	0.13	0.0020	0.19	-0.0003	0.13	0.0037	0.14	0.0023
68.15	0.8000	0.8240	0.01	0.0058	0.07	0.0024	0.14	0.0019	0.06	0.0031	0.07	0.0026
68.00	0.9000	0.9005	-0.02	-0.0007	0.05	-0.0002	0.11	0.0011	0.04	-0.0003	0.06	-0.0001
68.00	0.9500	0.9462	-0.03	-0.0027	0.03	-0.0013	0.07	0.0000	0.02	-0.0014	0.03	-0.0012
MEAN DEVIATION:			0.17	0.0061	0.10	0.0036	0.19	0.0049	0.10	0.0033	0.11	0.0033
MAX. DEVIATION:			0.53	0.0142	0.23	0.0094	0.42	0.0132	0.30	0.0067	0.24	0.0090

(1) ETHANOL C₂H₆O

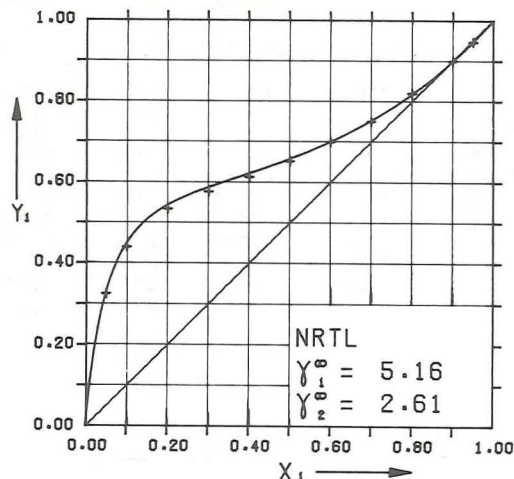
(2) WATER H₂O

+++++ ANTOINE CONSTANTS REGION +++++ CONSISTENCY
(1) 8.11220 1592.864 226.184 20- 93 C METHOD 1 +
(2) 8.07131 1730.630 233.426 1- 100 C METHOD 2 +

PRESSURE= 740.00 MM HG (0.987 BAR)

LIT: KIRSCHBAUM E., GERSTNER H., Z. VDI BEIH. VERFAHRENSTECHNIK 1, 10 (1939).

CONSTANTS: A12 A21 ALPHA12
MARGULES 1.6051 0.9067
VAN LAAR 1.6997 0.9681
WILSON 372.8463 950.1412
NRTL -85.7783 1303.4475 0.2928
UNIQUAC 72.8258 221.9193



EXPERIMENTAL DATA			MARGULES		VAN LAAR		WILSON		NRTL		UNIQUAC	
T DEG C	X1	Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1
89.75	0.0500	0.3240	-0.29	0.0030	0.07	-0.0053	0.26	-0.0084	-0.03	-0.0026	0.07	-0.0050
85.25	0.1000	0.4390	-0.43	-0.0085	-0.33	-0.0084	-0.47	-0.0017	-0.31	-0.0086	-0.33	-0.0078
82.10	0.2000	0.5340	0.23	-0.0178	-0.05	-0.0086	-0.42	0.0030	0.02	-0.0091	-0.08	-0.0071
80.70	0.3000	0.5770	0.36	-0.0201	0.03	-0.0111	-0.23	-0.0034	0.08	-0.0102	0.00	-0.0094
79.75	0.4000	0.6130	0.25	-0.0147	0.03	-0.0103	-0.10	-0.0078	0.05	-0.0082	0.01	-0.0089
78.90	0.5000	0.6520	0.05	-0.0065	-0.07	-0.0077	-0.10	-0.0090	-0.06	-0.0051	-0.08	-0.0066
78.25	0.6000	0.7000	-0.06	0.0031	-0.09	-0.0021	-0.06	-0.0051	-0.09	0.0003	-0.09	-0.0015
77.60	0.7000	0.7500	-0.27	0.0022	-0.25	-0.0039	-0.19	-0.0064	-0.25	-0.0021	-0.24	-0.0036
77.10	0.8000	0.8200	-0.49	0.0055	-0.43	0.0019	-0.35	0.0013	-0.44	0.0026	-0.42	0.0020
76.90	0.9000	0.8998	-0.61	0.0008	-0.53	0.0014	-0.47	0.0026	-0.54	0.0013	-0.53	0.0015
76.90	0.9500	0.9470	-0.65	-0.0007	-0.59	0.0009	-0.55	0.0021	-0.60	0.0006	-0.59	0.0009
MEAN DEVIATION:			0.33	0.0075	0.22	0.0056	0.29	0.0046	0.22	0.0046	0.22	0.0049
MAX. DEVIATION:			0.65	0.0201	0.59	0.0111	0.55	0.0090	0.60	0.0102	0.59	0.0094

(1) ETHANOL

C2H6O

(2) WATER

H2O

+++++ ANTOINE CONSTANTS REGION +++++ CONSISTENCY

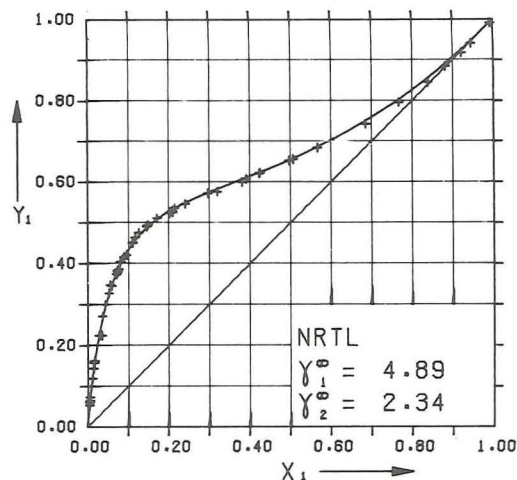
(1)	8.11220	1592.864	226.184	20-	93 C	METHOD 1	+
(2)	8.07131	1730.630	233.426	1-	100 C	METHOD 2	+

PRESSURE= 735.56 MM HG (0.981 BAR)

LIT: OTSUKI H., WILLIAMS F.C., CHEM. ENG. PROGR. SYMP. SER. 49, 55 (1953).

CONSTANTS: A12 A21 ALPHA12

MARGULES	1.5578	0.7726	
VAN LAAR	1.6139	0.8741	
WILSON	292.8677	919.7488	
NRTL	-153.7005	1337.2802	0.3047
UNIQUAC	-3.5341	263.8919	



EXPERIMENTAL DATA			MARGULES		VAN LAAR		WILSON		NRTL		UNIQUAC	
T DEG C	X1	Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1
98.50	0.0050	0.0528	0.68	0.0033	0.75	0.0009	0.78	-0.0001	0.72	0.0019	0.74	0.0011
98.40	0.0056	0.0597	0.72	0.0047	0.80	0.0021	0.83	0.0010	0.77	0.0032	0.79	0.0023
98.50	0.0065	0.0649	1.04	0.0018	1.12	-0.0011	1.16	-0.0024	1.09	0.0001	1.11	-0.0009
97.80	0.0067	0.0736	0.38	0.0087	0.47	0.0057	0.51	0.0045	0.43	0.0069	0.46	0.0060
96.60	0.0124	0.1190	0.44	0.0070	0.57	0.0029	0.63	0.0013	0.53	0.0043	0.57	0.0031
96.00	0.0155	0.1440	0.48	0.0091	0.62	0.0046	0.68	0.0030	0.58	0.0059	0.62	0.0048
95.40	0.0175	0.1580	0.27	0.0091	0.42	0.0046	0.48	0.0030	0.38	0.0057	0.41	0.0047
95.00	0.0180	0.1630	-0.04	0.0107	0.11	0.0061	0.17	0.0046	0.08	0.0073	0.11	0.0063
92.80	0.0292	0.2240	-0.30	0.0048	-0.15	0.0007	-0.10	-0.0001	-0.16	0.0010	-0.14	0.0006
92.00	0.0363	0.2250	-0.04	-0.0293	0.09	-0.0324	0.11	-0.0325	0.11	-0.0329	0.10	-0.0327
92.20	0.0374	0.2710	0.31	0.0117	0.44	0.0087	0.46	0.0087	0.46	0.0081	0.45	0.0085
90.90	0.0440	0.3000	-0.12	0.0127	-0.02	0.0108	-0.02	0.0116	0.02	0.0096	-0.00	0.0104
90.00	0.0538	0.3280	0.13	0.0048	0.17	0.0045	0.14	0.0064	0.25	0.0027	0.20	0.0041
89.70	0.0570	0.3480	0.17	0.0144	0.20	0.0146	0.15	0.0169	0.28	0.0125	0.22	0.0141
89.20	0.0607	0.3460	0.04	0.0009	0.05	0.0018	-0.00	0.0045	0.15	-0.0005	0.08	0.0013
88.20	0.0725	0.3740	0.12	-0.0034	0.06	-0.0006	-0.03	0.0033	0.18	-0.0033	0.09	-0.0011
87.50	0.0746	0.3830	-0.40	0.0004	-0.48	0.0035	-0.58	0.0076	-0.35	0.0008	-0.44	0.0030
87.90	0.0754	0.3800	0.06	-0.0045	-0.02	-0.0013	-0.12	0.0029	0.11	-0.0041	0.02	-0.0018
88.10	0.0782	0.3860	0.49	-0.0051	0.39	-0.0014	0.28	0.0030	0.52	-0.0043	0.43	-0.0019
87.50	0.0820	0.4050	0.18	0.0054	0.06	0.0097	-0.06	0.0143	0.20	0.0067	0.10	0.0092
86.90	0.0905	0.4170	0.18	-0.0001	0.02	0.0054	-0.13	0.0106	0.17	0.0023	0.05	0.0049
86.60	0.0970	0.4220	0.30	-0.0072	0.10	-0.0009	-0.06	0.0048	0.26	-0.0040	0.14	-0.0013
86.10	0.1110	0.4520	0.60	-0.0001	0.34	0.0078	0.15	0.0141	0.50	0.0048	0.37	0.0075
85.80	0.1120	0.4500	0.36	-0.0036	0.09	0.0044	-0.10	0.0108	0.25	0.0015	0.12	0.0042
85.40	0.1180	0.4640	0.26	0.0019	-0.04	0.0104	-0.23	0.0170	0.13	0.0076	-0.00	0.0102
85.30	0.1270	0.4760	0.58	0.0021	0.25	0.0114	0.04	0.0181	0.41	0.0087	0.28	0.0113
84.70	0.1470	0.4920	0.77	-0.0043	0.38	0.0063	0.17	0.0132	0.54	0.0041	0.41	0.0064
84.60	0.1470	0.4910	0.67	-0.0053	0.28	0.0053	0.07	0.0122	0.44	0.0031	0.31	0.0054
84.50	0.1490	0.4950	0.64	-0.0033	0.24	0.0074	0.03	0.0143	0.40	0.0053	0.27	0.0075
84.00	0.1720	0.5120	0.86	-0.0067	0.41	0.0048	0.21	0.0114	0.56	0.0034	0.44	0.0052

(CONT.)

83.30	0.2030	0.5240	0.92	-0.0164	0.44	-0.0047	0.26	0.0011	0.56	-0.0051	0.46	-0.0041
83.20	0.2090	0.5290	0.94	-0.0151	0.46	-0.0033	0.29	0.0022	0.57	-0.0035	0.48	-0.0027
83.00	0.2100	0.5260	0.76	-0.0186	0.28	-0.0070	0.11	-0.0015	0.39	-0.0071	0.30	-0.0063
83.20	0.2150	0.5360	1.06	-0.0115	0.57	0.0001	0.41	0.0054	0.68	0.0001	0.59	0.0008
82.80	0.2410	0.5460	1.11	-0.0150	0.63	-0.0041	0.50	0.0001	0.72	-0.0033	0.64	-0.0032
81.90	0.2970	0.5730	0.94	-0.0112	0.52	-0.0032	0.47	-0.0014	0.56	-0.0008	0.53	-0.0021
81.80	0.3200	0.5770	1.07	-0.0155	0.69	-0.0089	0.67	-0.0082	0.72	-0.0059	0.70	-0.0077
81.00	0.3810	0.6000	0.80	-0.0126	0.53	-0.0106	0.59	-0.0121	0.53	-0.0065	0.54	-0.0093
81.00	0.3910	0.6080	0.88	-0.0078	0.62	-0.0066	0.69	-0.0084	0.62	-0.0024	0.63	-0.0053
80.70	0.4230	0.6210	0.82	-0.0052	0.62	-0.0064	0.72	-0.0091	0.60	-0.0019	0.63	-0.0051
80.60	0.4250	0.6230	0.73	-0.0039	0.53	-0.0052	0.64	-0.0079	0.52	-0.0007	0.54	-0.0039
80.00	0.4970	0.6530	0.62	0.0007	0.53	-0.0053	0.68	-0.0090	0.50	-0.0005	0.54	-0.0041
80.10	0.5070	0.6560	0.79	-0.0001	0.71	-0.0067	0.87	-0.0105	0.67	-0.0019	0.72	-0.0055
79.50	0.5670	0.6840	0.56	0.0029	0.55	-0.0063	0.73	-0.0099	0.50	-0.0018	0.56	-0.0052
78.90	0.6850	0.7430	0.58	-0.0001	0.67	-0.0097	0.86	-0.0110	0.62	-0.0069	0.68	-0.0089
78.40	0.7670	0.7960	0.41	-0.0019	0.55	-0.0080	0.74	-0.0071	0.50	-0.0069	0.56	-0.0075
78.30	0.8380	0.8460	0.53	-0.0074	0.68	-0.0090	0.85	-0.0064	0.63	-0.0091	0.69	-0.0087
78.20	0.8810	0.8840	0.53	-0.0061	0.68	-0.0053	0.82	-0.0021	0.63	-0.0059	0.68	-0.0050
78.20	0.9210	0.9180	0.60	-0.0082	0.73	-0.0057	0.84	-0.0027	0.69	-0.0066	0.73	-0.0055
78.20	0.9440	0.9410	0.64	-0.0064	0.74	-0.0037	0.83	-0.0011	0.72	-0.0046	0.75	-0.0037
78.30	0.9930	0.9920	0.81	-0.0014	0.83	-0.0008	0.84	-0.0003	0.83	-0.0010	0.83	-0.0008
MEAN DEVIATION:			0.54	0.0070	0.43	0.0059	0.43	0.0074	0.47	0.0049	0.44	0.0056
MAX. DEVIATION:			1.11	0.0293	1.12	0.0325	1.16	0.0325	1.09	0.0329	1.11	0.0327

C₂H₆O

(1) ETHANOL

C2H6O

(2) WATER

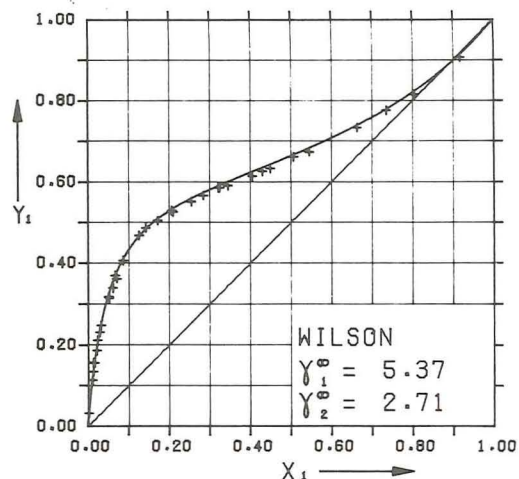
H2O

+++++ ANTOINE CONSTANTS REGION +++++ CONSISTENCY
 (1) 8.11220 1592.864 226.184 20- 93 C METHOD 1 +
 (2) 8.07131 1730.630 233.426 1- 100 C METHOD 2 +

PRESSURE= 760.00 MM HG (1.013 BAR)

LIT: RIEDER R.M., THOMPSON A.R., IND. ENG. CHEM. 41, 2905 (1949).

CONSTANTS: A12 A21 ALPHA12
 MARGULES 1.6170 0.7761
 VAN LAAR 1.6604 0.9171
 WILSON 288.9156 962.0073
 NRTL -136.0165 1351.8288 0.2968
 UNIQUAC 35.5784 242.5991



EXPERIMENTAL DATA			MARGULES		VAN LAAR		WILSON		NRTL		UNIQUAC	
T DEG C	X1	Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1
99.30	0.0028	0.0320	0.08	0.0018	0.11	0.0006	0.13	0.0000	0.09	0.0014	0.11	0.0008
96.90	0.0118	0.1130	-0.11	0.0003	-0.01	-0.0029	0.04	-0.0043	-0.06	-0.0012	-0.02	-0.0024
96.00	0.0137	0.1570	-0.59	0.0292	-0.48	0.0259	-0.44	0.0245	-0.54	0.0276	-0.50	0.0264
96.00	0.0144	0.1350	-0.44	0.0018	-0.33	-0.0015	-0.28	-0.0029	-0.38	0.0001	-0.34	-0.0011
95.60	0.0176	0.1560	-0.17	-0.0006	-0.06	-0.0040	-0.01	-0.0053	-0.11	-0.0025	-0.07	-0.0036
94.80	0.0222	0.1860	-0.08	-0.0013	0.03	-0.0046	0.08	-0.0056	-0.01	-0.0034	0.02	-0.0042
93.80	0.0246	0.2120	-0.65	0.0099	-0.53	0.0068	-0.49	0.0060	-0.57	0.0078	-0.54	0.0071
93.50	0.0302	0.2310	0.00	-0.0026	0.11	-0.0052	0.13	-0.0056	0.09	-0.0048	0.10	-0.0050
92.90	0.0331	0.2480	-0.14	-0.0005	-0.05	-0.0028	-0.02	-0.0028	-0.05	-0.0026	-0.05	-0.0027
90.50	0.0519	0.3180	-0.04	-0.0090	-0.02	-0.0086	-0.06	-0.0065	0.04	-0.0101	-0.01	-0.0088
90.50	0.0530	0.3140	0.09	-0.0168	0.10	-0.0162	0.05	-0.0140	0.16	-0.0178	0.11	-0.0165
89.40	0.0625	0.3390	-0.01	-0.0217	-0.05	-0.0198	-0.13	-0.0166	0.05	-0.0219	-0.03	-0.0201
88.40	0.0673	0.3700	-0.55	-0.0041	-0.61	-0.0015	-0.71	0.0021	-0.51	-0.0039	-0.59	-0.0019
88.60	0.0715	0.3620	0.03	-0.0231	-0.06	-0.0199	-0.17	-0.0158	0.05	-0.0224	-0.03	-0.0203
87.20	0.0871	0.4060	-0.15	-0.0140	-0.31	-0.0090	-0.46	-0.0037	-0.17	-0.0119	-0.28	-0.0094
85.40	0.1260	0.4680	0.23	-0.0128	-0.08	-0.0045	-0.29	0.0022	0.09	-0.0071	-0.04	-0.0047
84.50	0.1430	0.4870	0.02	-0.0125	-0.34	-0.0035	-0.55	0.0033	-0.17	-0.0058	-0.30	-0.0036
84.00	0.1720	0.5050	0.43	-0.0197	0.03	-0.0101	-0.17	-0.0039	0.18	-0.0116	0.07	-0.0100
83.40	0.2060	0.5300	0.61	-0.0168	0.21	-0.0076	0.04	-0.0024	0.33	-0.0079	0.24	-0.0072
83.00	0.2100	0.5270	0.29	-0.0220	-0.11	-0.0129	-0.28	-0.0079	0.00	-0.0131	-0.08	-0.0125
82.30	0.2550	0.5520	0.29	-0.0180	-0.06	-0.0108	-0.16	-0.0078	0.02	-0.0096	-0.04	-0.0102
82.00	0.2840	0.5670	0.34	-0.0140	0.02	-0.0085	-0.02	-0.0069	0.08	-0.0065	0.05	-0.0078
81.40	0.3210	0.5860	0.10	-0.0073	-0.14	-0.0043	-0.13	-0.0045	-0.11	-0.0014	-0.12	-0.0035
81.50	0.3240	0.5860	0.23	-0.0082	-0.01	-0.0054	0.01	-0.0057	0.02	-0.0025	0.01	-0.0046
81.20	0.3450	0.5910	0.11	-0.0096	-0.09	-0.0084	-0.04	-0.0096	-0.07	-0.0051	-0.07	-0.0076
80.90	0.4050	0.6140	0.25	-0.0046	0.16	-0.0078	0.29	-0.0112	0.16	-0.0037	0.19	-0.0070
80.50	0.4300	0.6260	0.02	-0.0003	-0.02	-0.0053	0.13	-0.0092	-0.03	-0.0009	0.01	-0.0045
80.20	0.4490	0.6330	-0.16	0.0006	-0.17	-0.0056	-0.00	-0.0099	-0.18	-0.0012	-0.14	-0.0048
80.00	0.5060	0.6610	-0.00	0.0087	0.07	-0.0006	0.27	-0.0054	0.05	0.0040	0.10	0.0003
79.50	0.5450	0.6730	-0.27	0.0054	-0.16	-0.0054	0.06	-0.0101	-0.19	-0.0009	-0.13	-0.0045

(CONT.)

78.80	0.6630	0.7330	-0.35	0.0071	-0.14	-0.0036	0.09	-0.0060	-0.18	-0.0005	-0.11	-0.0028
78.50	0.7350	0.7760	-0.36	0.0043	-0.11	-0.0030	0.12	-0.0030	-0.15	-0.0011	-0.08	-0.0022
78.40	0.8040	0.8150	-0.24	-0.0082	0.02	-0.0105	0.24	-0.0082	-0.02	-0.0100	0.05	-0.0098
78.30	0.9170	0.9060	-0.11	-0.0156	0.09	-0.0113	0.22	-0.0075	0.05	-0.0120	0.10	-0.0108

MEAN DEVIATION:	0.22	0.0098	0.14	0.0076	0.19	0.0071	0.15	0.0072	0.14	0.0073
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MAX. DEVIATION:	0.65	0.0292	0.61	0.0259	0.71	0.0245	0.57	0.0276	0.59	0.0264
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182
 C_2H_6O

(1) ETHANOL

C₂H₆O

(2) WATER

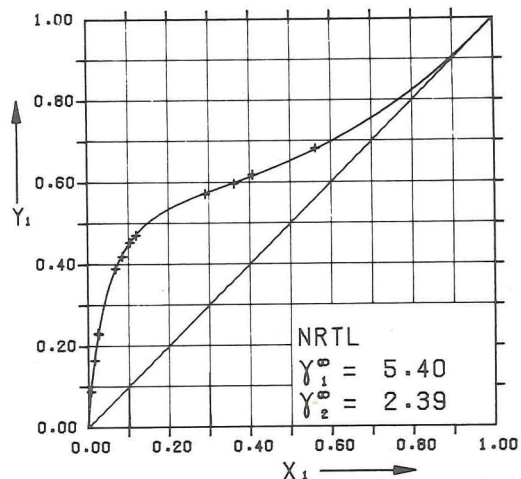
H₂O

***** ANTOINE CONSTANTS REGION ***** CONSISTENCY
(1) 8.11220 1592.864 226.184 20- 93 C METHOD 1 +
(2) 8.07131 1730.630 233.426 1- 100 C METHOD 2 +

PRESSURE= 706.00 MM HG (0.941 BAR)

LIT: RIUS A.,OTERO J.L.,MACARRON A.,CHEM.ENG.SCI.10,288(1959).

CONSTANTS: A12 A21 ALPHA12
MARGULES 1.6776 0.6791
VAN LAAR 1.7154 0.9110
WILSON 318.7353 963.8041
NRTL -170.3330 1426.5940 0.2967
UNIQUAC -27.8380 325.1825



EXPERIMENTAL DATA			MARGULES		VAN LAAR		WILSON		NRTL		UNIQUAC	
T DEG C	X1	Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1	DIFF T	DIFF Y1
95.90	0.0080	0.0880	0.16	0.0031	0.23	0.0008	0.26	-0.0001	0.19	0.0022	0.22	0.0012
93.80	0.0180	0.1650	0.34	-0.0021	0.44	-0.0048	0.48	-0.0058	0.40	-0.0037	0.43	-0.0046
91.80	0.0260	0.2320	-0.12	0.0128	-0.04	0.0106	-0.01	0.0101	-0.05	0.0109	-0.04	0.0106
91.60	0.0280	0.2290	0.02	-0.0018	0.10	-0.0037	0.13	-0.0040	0.10	-0.0036	0.11	-0.0038
86.40	0.0700	0.3900	-0.02	0.0003	-0.14	0.0041	-0.23	0.0076	-0.01	0.0014	-0.10	0.0034
85.60	0.0860	0.4190	0.42	-0.0061	0.23	-0.0006	0.11	0.0038	0.38	-0.0035	0.27	-0.0012
84.00	0.1050	0.4540	-0.02	-0.0035	-0.28	0.0035	-0.42	0.0085	-0.12	0.0008	-0.24	0.0031
83.40	0.1200	0.4710	0.11	-0.0067	-0.18	0.0010	-0.34	0.0062	-0.03	-0.0013	-0.15	0.0008
79.90	0.2910	0.5720	0.06	-0.0056	-0.06	-0.0053	0.04	-0.0071	-0.08	-0.0013	-0.07	-0.0037
79.40	0.3610	0.5970	0.10	-0.0006	0.15	-0.0065	0.36	-0.0112	0.09	-0.0012	0.14	-0.0047
78.80	0.4070	0.6180	-0.20	0.0073	-0.04	-0.0025	0.22	-0.0084	-0.13	0.0034	-0.05	-0.0008
77.70	0.5610	0.6820	-0.40	0.0149	-0.00	-0.0018	0.34	-0.0074	-0.11	0.0033	-0.01	-0.0005
MEAN DEVIATION:			0.17	0.0054	0.16	0.0038	0.24	0.0067	0.14	0.0030	0.15	0.0032
MAX. DEVIATION:			0.42	0.0149	0.44	0.0106	0.48	0.0112	0.40	0.0109	0.43	0.0106